

GLOBAL

The Crash of EgyptAir 990

Two years afterward the U.S. and Egyptian governments are still quarreling over the cause—a clash that grows out of cultural division, not factual uncertainty. A look at the flight data from a pilot's perspective, with the help of simulations of the accident, points to what the Egyptians must already know: the crash was caused not by any mechanical failure but by a pilot's intentional act

By William Langewiesche



Konstantin von Wedelstaedt/Wikimedia Commons

NOVEMBER 2001 ISSUE

SHARE 

SAVE 

I remember first hearing about the accident early in the morning after the airplane went down. It was October 31, 1999, Halloween morning. I was in my office when a fellow pilot, a former flying companion, phoned with the news: It was EgyptAir Flight 990, a giant twin-engine Boeing 767 on the way from New York to Cairo, with 217 people aboard. It had taken off from Kennedy Airport in the middle of the night, climbed to 33,000 feet, and flown normally for half an hour before mysteriously plummeting into the Atlantic Ocean sixty miles south of Nantucket. Rumor had it that the crew had said nothing to air-traffic control, that the flight had simply dropped off the New York radar screens. Soon afterward an outbound Air France flight had swung over the area, and had reported no fires in sight—only a dim and empty ocean far below. It was remotely possible that Flight 990 was still in the air somewhere, diverting toward a safe landing. But sometime around daybreak a Merchant Marine training ship spotted debris floating on the waves—aluminum scraps, cushions and clothing, some human remains. The midshipmen on board gagged from the stench of jet fuel—a planeload of unburned kerosene rising from shattered tanks on the ocean floor, about 250 feet below. By the time rescue ships and helicopters arrived, it was obvious that there would be no survivors. I remember reacting to the news with regret

for the dead, followed by a thought for the complexity of the investigation that now lay ahead. This accident had the markings of a tough case. The problem was not so much the scale of the carnage—a terrible consequence of the 767's size—but, rather, the still-sketchy profile of the upset that preceded it, this bewildering fall out of the sky on a calm night, without explanation, during an utterly uncritical phase of the flight.

I don't fly the 767, or any other airliner. In fact, I no longer fly for a living. But I know through long experience with flight that such machines are usually docile, and that steering them does not require the steady nerves and quick reflexes that passengers may imagine. Indeed, as we saw on September 11, steering them may not even require much in the way of training—the merest student-pilot level is

probably enough. It's not hard to understand why. Airplanes at their core are very simple devices—winged things that belong in the air. They are designed to be flyable, and they are. Specifically, the 767 has ordinary mechanical and hydraulic flight controls that provide the pilot with smooth and conventional responses; it is normally operated on autopilot, but can easily be flown by hand; if you remove your hands from the controls entirely, the airplane sails on as before, until it perhaps wanders a bit, dips a wing, and starts into a gentle descent; if you pull the nose up or push it down (within reason) and then fold your arms, the airplane returns unassisted to steady flight; if you idle the engines, or shut them off entirely, the airplane becomes a rather well behaved glider. It has excellent forward visibility, through big windshields. It has a minimalist cockpit that may look complicated to the untrained eye but is a masterpiece of clean design. It can easily be managed by the standard two-person crew, or even by one pilot alone. The biggest problem in flying the airplane on a

From Atlantic Unbound:

Interviews: "Culture Crash"
(November 15, 2001)

A conversation with William
Langewiesche on the cultural
reverberations of a seemingly
straightforward airplane crash

routine basis is boredom. Settled into the deep sky at 33,000 feet, above the weather and far from any obstacle, the 767 simply makes very few demands.

Not that it's idiot-proof, or necessarily always benign. As with any fast and heavy airplane, operating a 767 safely even under ordinary circumstances requires anticipation, mental clarity, and a practical understanding of the various systems. Furthermore, when circumstances are *not* ordinary—for example, during an engine failure just after takeoff or an encounter with unexpected wind shear during an approach to landing—a wilder side to the airplane's personality suddenly emerges. Maintaining control then requires firm action and sometimes a strong arm. There's nothing surprising about this: all airplanes misbehave on occasion, and have to be disciplined. "Kicking the dog," I called it in the ornery old cargo crates I flew when I was in college—it was a regular part of survival. In the cockpits of modern jets it is rarely necessary. Nonetheless, when trouble occurs in a machine as massive and aerodynamically slick as the 767, if it is not quickly suppressed the consequences can blossom out of control. During a full-blown upset like that experienced by the Egyptian crew, the airplane may dive so far past its tested limits that it exceeds the very scale of known engineering data—falling off the graphs as well as out of the sky. Afterward the profile can possibly be reconstructed mathematically by aerodynamicists and their like, but it cannot be even approximated by pilots in flight if they expect to come home alive.

I got a feel for the 767's dangerous side last summer, after following the accident's trail from Washington, D.C., to Cairo to the airplane's birthplace, in Seattle, where Boeing engineers let me fly a specially rigged 767 simulator through a series of relevant upsets and recoveries along with some sobering replays of Flight 990's final moments. These simulations had been flown by investigators more than a year before and had been reported on in detail in the publicly released files. Boeing's argument was not that the 767 is a flawless design but, more narrowly, that none of the imaginable failures of its flight-control systems could explain the known facts of this accident.

But that's getting ahead of the story. Back on October 31, 1999, with the first news of the crash, it was hard to imagine any form of pilot error that could have condemned the airplane to such a sustained and precipitous dive. What switch could the crew have thrown, what lever? Nothing came to mind. And why had they perished so silently, without a single distress call on the radio? A total electrical failure was very unlikely, and would not explain the loss of control. A fire would have given them time to talk. One thing was certain: the pilots were either extremely busy or incapacitated from the start. Of course there was always the possibility of a terrorist attack—a simple if frightening solution. But otherwise something had gone terribly wrong with the airplane itself, and that could be just as bad. There are more than 800 Boeing 767s in the world's airline fleet, and they account for more transatlantic flights than all other airplanes combined. They are also very similar in design to the smaller and equally numerous Boeing 757s. So there was plenty of reason for alarm.

"EgyptAir Nine-ninety, Radio Contact Lost"

One of the world's really important divides lies between nations that react well to accidents and nations that do not. This is as true for a confined and technical event like the crash of a single flight as it is for political or military disasters. The first requirement is a matter of national will, and never a sure thing: it is the intention to get the story right, wherever the blame may lie. The second requirement follows immediately upon the first, and is probably easier to achieve: it is the need for people in the aftermath to maintain even tempers and open minds. The path they follow may not be simple, but it can provide for at least the possibility of effective resolutions.

In the case of EgyptAir Flight 990 the only information available at first was external. The airplane had arrived in New York late on a flight from Los Angeles, and had paused to refuel, take on passengers, and swap crews. Because of the scheduled duration of the flight to Cairo, two cockpit crews had been assigned to the ocean crossing—an "active crew," including the aircraft commander, to handle the first and

last hours of the flight; and a "cruise crew," whose role was essentially to monitor the autopilot during the long, sleepy mid-Atlantic stretch. Just before midnight these four pilots rode out to the airport on a shuttle bus from Manhattan's Pennsylvania Hotel, a large establishment where EgyptAir retained rooms for the use of its personnel. The pilots had been there for several days and, as usual, were well rested. Also in the bus was one of the most senior of EgyptAir's captains, the company's chief 767 pilot, who was not scheduled to fly but would be "deadheading" home to Cairo. An EgyptAir dispatcher rode out on the bus with them, and subsequently reported that the crew members looked and sounded normal. At the airport he gave them a standard briefing and an update on the New York surface weather, which was stagnant under a low, thin overcast, with light winds and thickening haze.

Flight 990 pushed back from the gate and taxied toward the active runway at 1:12 A.M. Because there was little other traffic at the airport, communications with the control tower were noticeably relaxed. At 1:20 Flight 990 lifted off. It topped the clouds at 1,000 feet and turned out over the ocean toward a half moon rising above the horizon. The airplane was identified and tracked by air-traffic-control radar as it climbed through the various New York departure sectors and entered the larger airspace belonging to the en-route controllers of New York Center; its transponder target and data block moved steadily across the controllers' computer-generated displays, and its radio transmissions sounded perhaps a little awkward, but routine. At 1:44 it leveled off at the assigned 33,000 feet.

The en-route controller working the flight was a woman named Ann Brennan, a private pilot with eight years on the job. She had the swagger of a good controller, a real pro. Later she characterized the air traffic that night as slow, which it was—during the critical hour she had handled only three other flights. The offshore military-exercise zones, known as warning areas, were inactive. The sky was sleeping.

At 1:47 Brennan said, "EgyptAir Nine-ninety, change to my frequency one-two-five-point-niner-two."

EgyptAir acknowledged the request with a friendly "Good day," and after a pause checked in on the new frequency: "New York, EgyptAir Nine-nine-zero heavy, good

morning."

Brennan answered, "EgyptAir Nine-ninety, roger."

That was the last exchange. Brennan noticed that the flight still had about fifteen minutes to go before leaving her sector. Wearing her headset, she stood up and walked six feet away to sort some paperwork. A few minutes later she approved a request by Washington Center to steer an Air France 747 through a corner of her airspace. She chatted for a while with her supervisor, a man named Ray Redhead. In total she spent maybe six minutes away from her station, a reasonable interval on such a night. It was just unlucky that while her back was turned Flight 990 went down.

A computer captured what she would have seen—a strangely abstract death no more dramatic than a video game. About two minutes after the final radio call, at 1:49:53 in the morning, the radar swept across EgyptAir's transponder at 33,000 feet. Afterward, at successive twelve-second intervals, the radar read 31,500, 25,400, and 18,300 feet—a descent rate so great that the air-traffic-control computers interpreted the information as false, and showed "XXXX" for the altitude on Brennan's display. With the next sweep the radar lost the transponder entirely, and picked up only an unenhanced "primary" blip, a return from the airplane's metal mass. The surprise is that the radar continued to receive such returns (which show only location, and not altitude) for nearly another minute and a half, indicating that the dive must have dramatically slowed or stopped, and that the 767 remained airborne, however tenuously, during that interval. A minute and a half is a long time. As the Boeing simulations later showed, it must have been a strange and dreamlike period for the pilots, hurtling through the night with no chance of awakening.

When radar contact was lost, the display for EgyptAir 990 began to "coast," indicating that the computers could no longer find a correlation between the stored flight plan and the radar view of the sky. When Brennan noticed, she stayed cool. She said, "EgyptAir Nine-ninety, radar contact lost, recycle transponder, squawk one-seven-one-two." EgyptAir did not answer, so she tried again at unhurried intervals over the following ten minutes. She advised Ray Redhead of the problem, and he passed the word along. She called an air-defense radar facility, and other air-traffic-

control centers as far away as Canada, to see if by any chance someone was in contact with the flight. She asked a Lufthansa crew to try transmitting to EgyptAir from up high. Eventually she brought in Air France for the overflight. The prognosis was of course increasingly grim, but she maintained her professional calm. She continued to handle normal operations in her sector while simultaneously setting the search-and-rescue forces in motion. Half an hour into the process, when a controller at Boston Center called and asked, "Any luck with the EgyptAir?" she answered simply, "No."

Government Lite Meets Government Heavy

Among the dead were 100 Americans, eighty-nine Egyptians (including thirty-three army officers), twenty-two Canadians, and a few people of other nationalities. As the news of the disaster spread, hundreds of frantic friends and relatives gathered at the airports in Los Angeles, New York, and Cairo. EgyptAir officials struggled to meet people's needs—which were largely, of course, for the sort of information that no one yet had. Most of the bodies remained in and around the wreckage at the bottom of the sea. Decisions now had to be made, and fast, about the recovery operation and the related problem of an investigation. Because the airplane had crashed in international waters, Egypt had the right to lead the show. Realistically, though, it did not have the resources to salvage a heavy airplane in waters 250 feet deep and 5,000 miles away.

The solution was obvious, and it came in the form of a call to the White House from Egyptian President Hosni Mubarak, an experienced military pilot with close ties to EgyptAir, requesting that the investigation be taken over by the U.S. government. The White House in turn called Jim Hall, the chairman of the National Transportation Safety Board, an investigative agency with a merited reputation for competence. Hall, a Tennessee lawyer and friend of the Gores, had in the aftermath of the TWA Flight 800 explosion parlayed his position into one of considerable visibility. The Egyptians

produced a letter formally signing over the investigation to the United States, an option accorded under international convention, which would place them in a greatly diminished role (as "accredited representatives") but would also save them trouble and money. Mubarak is said to have regretted the move ever since.

In retrospect it seems inevitable that the two sides would have trouble getting along. The NTSB is a puritanical construct, a small federal agency without regulatory power whose sole purpose is to investigate accidents and issue safety recommendations that might add to the public discourse. Established in 1967 as an "independent" unit of the Washington bureaucracy, and shielded by design from the political currents of that city, the agency represents the most progressive American thinking on the role and character of good government. On call twenty-four hours a day, with technical teams ready to travel at a moment's notice, it operates on an annual baseline budget of merely \$62 million or so, and employs only about 420 people, most of whom work at the headquarters on four floors of Washington's bright and modern Loews L'Enfant Plaza Hotel. In part because the NTSB seems so lean, and in part because by its very definition it advocates for the "right" causes, it receives almost universally positive press coverage. The NTSB is technocratic. It is clean. It is Government Lite.

EgyptAir, in contrast, is Government Heavy—a state-owned airline with about 600 pilots and a mixed fleet of about forty Boeings and Airbuses that serves more than eighty destinations worldwide and employs 22,000 people. It operates out of dusty Stalinist-style office buildings at the Cairo airport, under the supervision of the Ministry of Transport, from which it is often practically indistinguishable. It is probably a safe airline, but passengers dislike it for its delays and shoddy service. They call it Air Misère, probably a play on the airline's former name, Misr Air ("Misr" is Arabic for "Egypt"). It has been treated as a fiefdom for years by Mubarak's old and unassailable air-force friends, and particularly by the company's chairman, a man named Mohamed Fahim Rayan, who fights off all attempts at reform or privatization. This is hardly a secret. In parliamentary testimony six months before the crash of Flight 990, Rayan said, "My market is like a water pond which I developed over the years. It is quite unreasonable for alien people to come and seek to catch fish in my pond." His critics answer that the pond is stagnant and stinks of corruption—but

this, too, is nothing new. The greatest pyramids in Egypt are made not of stone but of people: they are the vast bureaucracies that constitute society's core, and they function not necessarily to get the "job" done but to reward the personal loyalty of those at the bottom to those at the top. Once you understand that, much of the rest begins to make sense. The bureaucracies serve mostly to shelter their workers and give them something like a decent life. They also help to define Cairo. It is a great capital city, as worldly as Washington, D.C., and culturally very far away.

An official delegation traveled from Cairo to the United States and ended up staying for more than a year. It was led by two EgyptAir pilots, Mohsen al-Missiry, an experienced accident investigator on temporary assignment to the Egyptian Civil Aviation Authority for this case, and Shaker Kelada, who had retired from active flying to become a flight-operations manager and eventually vice-president for safety and quality assurance. These men were smart and tough, and managed a team primarily of EgyptAir engineers, many of whom were very sharp.

The U.S. Navy was given the job of salvage, and it in turn hired a contractor named Oceaneering, which arrived with a ship and grapples and remote-controlled submarines. The debris was plotted by sonar, and found to lie in two clusters: the small "west field," which included the left engine; and, 1,200 feet beyond it in the direction of flight, the "east field," where most of the airplane lay. From what was known of the radar profile and from the tight concentration of the debris, it began to seem unlikely that an in-flight explosion was to blame. The NTSB said nothing. Nine days after the accident the flight-data recorder—the "black box" that records flight and systems data—was retrieved and sent to the NTSB laboratory in Washington. The NTSB stated tersely that there was preliminary evidence that the initial dive may have been a "controlled descent." Five days later, on Sunday, November 14, a senior official at the Egyptian Transportation Ministry—an air-force general and a former EgyptAir pilot—held a news conference in Cairo and, with Rayan at his side, announced that the evidence from the flight-data recorder had been inconclusive but the dive could be explained only by a bomb in the cockpit or in the lavatory directly behind it. It was an odd assertion to make, but of little importance, because the second black box, the cockpit voice recorder, had been salvaged the night before and

was sent on Sunday to the NTSB. The tape was cleaned and processed, and a small group that included a translator (who was not Egyptian) gathered in a listening room at L'Enfant Plaza to hear it through.

"I Rely on God"

Listening to cockpit recordings is a tough and voyeuristic duty, restricted to the principal investigators and people with specific knowledge of the airplane or the pilots, who might help to prepare an accurate transcript. Experienced investigators grow accustomed to the job, but I talked to several who had heard the EgyptAir tape, and they admitted that they had been taken aback. Black boxes are such pitiless, unblinking devices. When the information they contained from Flight 990 was combined with the radar profile and the first, sketchy information on the crew, this was the story it seemed to tell:

The flight lasted thirty-one minutes. During the departure from New York it was captained, as required, by the aircraft commander, a portly senior pilot named Ahmad al-Habashi, fifty-seven, who had flown thirty-six years for the airline. Habashi of course sat in the left seat. In the right seat was the most junior member of the crew, a thirty-six-year-old co-pilot who was progressing well in his career and looking forward to getting married. Before takeoff the co-pilot advised the flight attendants by saying, in Arabic, "In the name of God, the merciful, the compassionate. Cabin crew takeoff position." This was not unusual.

After takeoff the autopilot did the flying. Habashi and the co-pilot kept watch, talked to air-traffic control, and gossiped about their work. The cockpit door was unlocked, which was fairly standard on EgyptAir flights. Various flight attendants came in and left; for a while the chief pilot, the man who was deadheading back to Cairo, stopped by the cockpit to chat. Then, twenty minutes into the flight, the "cruise" co-pilot, Gameel al-Batouti, arrived. Batouti was a big, friendly guy with a reputation for telling jokes and enjoying life. Three months short of sixty, and mandatory

retirement, he was unusually old for a co-pilot. He had joined the airline in his mid-forties, after a career as a flight instructor for the air force, and had rejected several opportunities for command. His lack of ambition was odd but not unheard of: his English was poor and might have given him trouble on the necessary exams; moreover, as the company's senior 767 co-pilot, he made adequate money and had his pick of long-distance flights. Now he used his seniority to urge the junior co-pilot to cede the right seat ahead of the scheduled crew change. When the junior man resisted, Batouti said, "You mean you're not going to get up? You will get up. Go and get some rest and come back." The junior co-pilot stayed in his seat a bit longer and then left the cockpit. Batouti took the seat and buckled in.

Batouti was married and had five children. Four of them were grown and doing well. His fifth child was a girl, age ten, who was sick with lupus but responding to treatment that he had arranged for her to receive in Los Angeles. Batouti had a nice house in Cairo. He had a vacation house on the beach. He did not drink heavily. He was moderately religious. He had his retirement planned. He had acquired an automobile tire in New Jersey the day before, and was bringing it home in the cargo hold. He had also picked up some free samples of Viagra, to distribute as gifts.

Captain Habashi was more religious, and was known to pray sometimes in the cockpit. He and Batouti were old friends. Using Batouti's nickname, he said, in Arabic, "How are you, Jimmy?" They groused to each other about the chief pilot and about a clique of young and arrogant "kids," junior EgyptAir pilots who were likewise catching a ride back to the Cairo base. One of those pilots came into the cockpit dressed in street clothes. Habashi said, "What's with you? Why did you get all dressed in red like that?" Presumably the man then left. Batouti had a meal. A female flight attendant came in and offered more. Batouti said pleasantly, "No, thank you, it was marvelous." She took his tray.

At 1:47 A.M. the last calls came in from air-traffic control, from Ann Brennan, far off in the night at her display. Captain Habashi handled the calls. He said, "New York, EgyptAir Nine-nine-zero heavy, good morning," and she answered with her final "EgyptAir Nine-ninety, roger."

At 1:48 Batouti found the junior co-pilot's pen and handed it across to Habashi. He said, "Look, here's the new first officer's pen. Give it to him, please. God spare you." He added, "To make sure it doesn't get lost."

Habashi said, "Excuse me, Jimmy, while I take a quick trip to the toilet." He ran his electric seat back with a whir. There was the sound of the cockpit door moving.

Batouti said, "Go ahead, please."

Habashi said, "Before it gets crowded. While they are eating. And I'll be back to you."

Again the cockpit door moved. There was a *clunk*. There was a *clink*. It seems that Batouti was now alone in the cockpit. The 767 was at 33,000 feet, cruising peacefully eastward at .79 Mach.

At 1:48:30 a strange, wordlike sound was uttered, three syllables with emphasis on the second, perhaps more English than Arabic, and variously heard on the tape as "control it," "hydraulic," or something unintelligible. The NTSB ran extensive speech and sound-spectrum studies on it, and was never able to assign it conclusively to Batouti or to anyone else. But what is clear is that Batouti then softly said, "*Tawakkalt ala Allah*," which proved difficult to translate, and was at first rendered incorrectly, but essentially means "I rely on God." An electric seat whirred. The autopilot disengaged, and the airplane sailed on as before for another four seconds. Again Batouti said, "I rely on God." Then two things happened almost simultaneously, according to the flight-data recorder: the throttles in the cockpit moved back fast to minimum idle, and a second later, back at the tail, the airplane's massive elevators (the pitch-control surfaces) dropped to a three-degrees-down position. When the elevators drop, the tail goes up; and when the tail goes up, the nose points down. Apparently Batouti had chopped the power and pushed the control yoke forward.

The effect was dramatic. The airplane began to dive steeply, dropping its nose so quickly that the environment inside plunged to nearly zero gs, the weightless condition of space. Six times in quick succession Batouti repeated, "I rely on God." His tone was calm. There was a loud thump. As the nose continued to pitch

downward, the airplane went into the negative-g range, nudging loose objects against the ceiling. The elevators moved even farther down. Batouti said, "I rely on God."

Somehow, in the midst of this, now sixteen seconds into the dive, Captain Habashi made his way back from the toilet. He yelled, "What's happening? What's happening?"

Batouti said, "I rely on God."

The wind outside was roaring. The airplane was dropping through 30,800 feet, and accelerating beyond its maximum operating speed of .86 Mach. In the cockpit the altimeters were spinning like cartoon clocks. Warning horns were sounding, warning lights were flashing—low oil pressure on the left engine, and then on the right. The master alarm went off, a loud high-to-low warble.

For the last time Batouti said, "I rely on God."

Again Habashi shouted, "What's happening?" By then he must have reached the left control yoke. The negative gs ended as he countered the pitch-over, slowing the rate at which the nose was dropping. But the 767 was still angled down steeply, 40 degrees below the horizon, and it was accelerating. The rate of descent hit 39,000 feet a minute.

"What's happening, Gameel? What's happening?"

Habashi was clearly pulling very hard on his control yoke, trying desperately to raise the nose. Even so, thirty seconds into the dive, at 22,200 feet, the airplane hit the speed of sound, at which it was certainly not meant to fly. Many things happened in quick succession in the cockpit. Batouti reached over and shut off the fuel, killing both engines. Habashi screamed, "What is this? What is this? Did you shut the engines?" The throttles were pushed full forward—for no obvious reason, since the engines were dead. The speed-brake handle was then pulled, deploying drag devices on the wings.

At the same time, there was an unusual occurrence back at the tail: the right-side and left-side elevators, which normally move together to control the airplane's pitch, began to "split," or move in opposite directions. Specifically: the elevator on the right remained down, while the left-side elevator moved up to a healthy recovery position. That this could happen at all was the result of a design feature meant to allow either pilot to overpower a mechanical jam and control the airplane with only one elevator. The details are complex, but the essence in this case seemed to be that the right elevator was being pushed down by Batouti while the left elevator was being pulled up by the captain. The NTSB concluded that a "force fight" had broken out in the cockpit.

Words were failing Habashi. He yelled, "Get away in the engines!" And then, incredulously, "... shut the engines!"

Batouti said calmly, "It's shut."

Habashi did not have time to make sense of the happenings. He probably did not have time to get into his seat and slide it forward. He must have been standing in the cockpit, leaning over the seatback and hauling on the controls. The commotion was horrendous. He was reacting instinctively as a pilot, yelling, "Pull!" and then, "Pull with me! Pull with me! Pull with me!"

It was the last instant captured by the on-board flight recorders. The elevators were split, with the one on the right side, Batouti's side, still pushed into a nose-down position. The ailerons on both wings had assumed a strange upswept position, normally never seen on an airplane. The 767 was at 16,416 feet, doing 527 miles an hour, and pulling a moderately heavy 2.4 gs, indicating that the nose, though still below the horizon, was rising fast, and that Habashi's efforts on the left side were having an effect. A belated recovery was under way. At that point, because the engines had been cut, all nonessential electrical devices were lost, blacking out not only the recorders, which rely on primary power, but also most of the instrument displays and lights. The pilots were left to the darkness of the sky, whether to work together or to fight. I've often wondered what happened between those two men during the 114 seconds that remained of their lives. We'll never know. Radar reconstruction showed

that the 767 recovered from the dive at 16,000 feet and, like a great wounded glider, soared steeply back to 24,000 feet, turned to the southeast while beginning to break apart, and shed its useless left engine and some of its skin before giving up for good and diving to its death at high speed.

Conflicting Realities

When this evidence emerged at the NTSB, the American investigators were shocked but also relieved by the obvious conclusion. There was no bomb here. Despite initial fears, there was nothing wrong with the airplane. The apparent cause was pilot error at its extreme: Batouti had gone haywire. Every detail that emerged from the two flight recorders fit that scenario: the sequence of the switches and controls that were moved, the responses of the airplane, and the words that were spoken, however cryptic and incomplete. Batouti had waited to be alone in the cockpit, and had intentionally pushed the airplane to its death. He had even fought the captain's valiant attempt at recovery. Why? Professionally, the NTSB didn't need to care. It was up to the criminal investigators at the FBI to discover if this was a political act, or the result of a plot. Even at the time, just weeks after the airplane went down, it was hard to imagine that Batouti had any terrorist connections, and indeed, the FBI never found any such evidence. But in pure aviation terms it didn't really matter why Batouti did it, and pure aviation is what the NTSB is all about. So this was easy—Crash Investigation 101. The guy to blame was dead. The NTSB wouldn't have to go after Boeing—a necessary task on occasion, but never a pleasant prospect. The wreckage, which was still being pulled out of the ocean, would not require tedious inspection. The report could be written quickly and filed away, and the NTSB could move on to the backlog of work that might actually affect the future safety of the flying public.

When Jim Hall, the NTSB chairman, held a news conference to address the initial findings, on November 19, 1999, he was culturally sensitive, responsible, and very strict about the need to maintain an open mind. There had been leaks to the press about the content of the cockpit voice recorder. It was being said that Batouti's behavior had been strange during the dive and that he had recited Muslim prayers. Hall scolded the assembled reporters for using unofficial information and exciting the public's emotions. He made a show of being careful with his own choice of words. He said that the accident "might, and I emphasize *might*, be the result of a deliberate act." He did not say "suicide" or "Arab" or "Muslim." He did not even say "Batouti." He said, "No one wants to get to the bottom of this mystery quicker than those investigating this accident, both here and in Egypt, but we won't get there on a road paved with leaks, supposition, speculation, and spin. That road does not lead to the truth, and the truth is what both the American people and the Egyptian people seek." It was standard stuff, a prelude to a quick wrapping up of the investigation. The Egyptian delegation, which had moved into rooms at the Loews L'Enfant Plaza Hotel, might have felt grateful to have such a man at the NTSB to guide them through these difficult times. Instead the Egyptians were outraged.

At the NTSB this came as a surprise. Looking back, it's possible to see signs of a disconnect, especially the Egyptian government's baffling speculation about a bomb in the forward lavatory; but just the day before Hall's press conference the Egyptian ambassador had heaped praise on the NTSB and the investigation. Now, suddenly and with startling vigor, the Egyptian delegation went on the offensive. The leader of the charge, Shaker Kelada, later told me about running across one of the American investigators in the halls of the NTSB. When the investigator mentioned with satisfaction that the work might wrap up within a few weeks, Kelada brought him up short with the news that he'd better change his plans—because far from being over, the investigation had hardly begun.

First the Egyptians had to prepare the ground: the delegation started to loudly criticize the performance and intentions of Boeing, the FBI, and the entire NTSB. Kelada said that Batouti was the scapegoat, and that this was happening because it was an *Egyptian* airliner that had gone down. It did not escape Kelada's attention that

the legendary head of aviation investigations at the NTSB—a brilliant and abrasive engineer named Bernard Loeb, who was overseeing the Flight 990 inquiry—was Jewish and something of a Zionist.

Loeb retired last spring; Kelada implied to me last summer that this was a deception, and that Loeb continued to pull the strings. Loeb laughed when I mentioned it to him afterward. He was looking forward to spending time with his grandchildren. But at the same time, he was angry that Egypt, after receiving \$1.3 billion in American assistance every year, would have used any of its budget to cause the United States unnecessary expense by prolonging an investigation that for the NTSB alone had so far cost \$17 million. As to Zionism, Loeb did seem bothered by aspects of the Egyptian culture. I got the feeling, though, that his opinion was fresh—that it stemmed from his contacts with EgyptAir, rather than from experiences that had preceded them.

But it didn't really matter who at the NTSB was in charge of the investigation. In faraway Cairo, inevitably, it was seen as unfair. From the day that Flight 990's recorder tape was transcribed and word of its contents began to leak out, the feeling in Egypt was that all Arabs were under attack, and that the assault had been planned. More than a year after the crash I met a sharp young reporter in Cairo who continued to seethe about it. He said, "For many Egyptians it was a big example of this business of dictating the reality. What made many people question the authenticity of the U.S. claims was the rush to conclusions ... The rush, the interpretation of a few words, it left no chance. The whole thing seemed to apply within a framework of an American sort of soap opera, one of those movies you make. You know—this is a fanatic, he comes from the Middle East, he utters a few religious words, he brings the plane down." But what if Batouti really had brought the plane down—where did the reporter's reaction leave Egypt? Earlier the reporter had written critically about the corruption at EgyptAir, but he refused even to think critically about it anymore.

The reporter's anger was similar, at least superficially, to the anger that was seething through Shaker Kelada and the rest of the Egyptian delegation in November of 1999. For Jim Hall, Bernard Loeb, and others at the NTSB, the source of the problem

seemed at first to be the media coverage, which was typically overeager. Rumors of suicide had circulated in the press almost since the airplane hit the water, but it was only after the voice recorder was recovered that the reports began to make uninformed reference to Muslim prayers. Three days before Hall's press conference *The Washington Post* ran a headline saying, "PILOT PRAYED, THEN SHUT OFF JET'S AUTOPILOT." Television stations speculated that the "prayer" was the *shahada* ("There is no god but God; Muhammad is the messenger of God"), as if this were what one might say before slaughtering infidels. When the actual Arabic words—*Tawakkalt ala Allah*—became public, some news outlets gave the following translation: "I have made my decision. I put my fate in God's hands." This was reported so widely that the NTSB took the unusual step of announcing that "I have made my decision" had never been spoken. By implication, "I place my fate ..." had.

When NTSB investigators explained their lack of control over the American press, the Egyptians scoffed and pointed out—correctly—that the reporters' sources were people inside the investigation. And anyway, the Egyptians added, what Batouti had said was not "I put my fate in God's hands"—as the NTSB's interpreter had claimed—but, rather, "I rely on God." The investigators blinked at the subtlety of this distinction, and made the necessary changes to the transcript. Then the Egyptians produced a letter from an Islamic scholar in Cairo who certified that the meaning of *Tawakkalt ala Allah* is "I depend in my daily affairs on the omnipotent Allah alone." The Egyptians wanted the letter inserted into the record, but were willing to allow "I rely on God" to remain in the transcript. Again, the investigators blinked. This was not the sort of thing they normally dealt with. They tried sometimes to bridge the gap as they might have with Americans, with a nudge and a smile, but it got them nowhere.

In essence the Egyptians were making two intertwined arguments: first, that it was culturally impossible for Batouti to have done what the NTSB believed; second, that the NTSB lacked the cultural sensitivity to understand what was on the cockpit voice recorder. With those arguments as a starting point, the Egyptians tore into the complexities of the evidence, disputing any assumptions or conclusions the NTSB put forward and raising new questions at every possible turn—a process that continues to this day. They were tenacious. For example (and this is just a small sample of the

Egyptians' arguments): When Batouti said "*Tawakkalt ala Allah*," he was not preparing to die but responding in surprise to something wrong with the flight. He said it quietly, yes, but with emotion that the Americans lacked the cultural sensitivity to hear. When he started the dive, he was trying to avoid a plane or a missile outside. If not that, then the airplane went into the dive on its own. When he idled the engines, it was to keep from gaining speed. When he cut the engines, he was going through the required restart procedure, because he erroneously believed—on the basis of the low-oil-pressure warning light that flashed in the cockpit—that the engines had flamed out. Apparently Habashi made the same mistake, which is why he discussed engine cuts. When Habashi called "Pull with me!," Batouti did exactly that. The split elevators were like the upswept ailerons—either an aerodynamic anomaly, resulting from the unknown pressures of ultra-high-speed flight on the 767, or, more simply, an error in the flight-data recorder. Whichever way, the Egyptians argued that expensive wind-tunnel testing was necessary at high Mach numbers near the speed of sound.

Meanwhile, most of the wreckage had been recovered and spread out in a hangar in Rhode Island. A second salvage operation was mounted in the spring to coincide with a state visit by Mubarak to Washington. It went to the west debris field and brought up the left engine and a boatload of worthless scraps. At the NTSB a story circulated about Al Gore, who was said to have angered Mubarak by making a casual reference to "the suicide flight." There was a short flap about that. The investigation continued. The documentation grew. The possibilities multiplied and ran off in a hundred directions. An airline pilot observing the scene said to me, "It could have been this, it could have been that. Bottom line is, it could have been anything except their guy."

The Search for a Motive

While the Egyptians were proposing theory after theory to absolve Batouti, the FBI was conducting a criminal investigation, collecting evidence that provided for his possible motive. Mostly through interviews with employees of the Pennsylvania Hotel, the FBI found that Batouti had a reputation for sexual impropriety—and not merely by the prudish standards of America. It

was reported that on multiple occasions over the previous two years he had been suspected of exposing himself to teenage girls, masturbating in public, following female guests to their rooms, and listening at their doors. Some of the maids, it was said, were afraid of him, and the hotel security guards had once brought him in for questioning and a warning. Apparently the hotel had considered banning him. The FBI learned that EgyptAir was aware of these problems and had warned Batouti to control his behavior. He was not considered to be a dangerous man—and certainly he was more sad than bad. In fact, there was a good side to Batouti that came out in these interviews as well. He was very human. Many people were fond of him, even at the hotel.

But a story soon surfaced that an altercation may have occurred during the New York layover before the fatal departure. The FBI was told that there had been trouble, and possibly an argument with the chief pilot, who was also staying at the hotel. It was hypothesized that the chief pilot might have threatened disciplinary action upon arrival back in Cairo—despite the public humiliation that would entail. Was that perhaps Batouti's motive? Did the killing of 217 people result from a simple act of vengeance against one man? The evidence was shaky at best. Then, in February of 2000, an EgyptAir pilot named Hamdi Hanafi Taha, forty-nine, landed in England and requested political asylum, claiming that he had information on the accident. FBI and NTSB investigators flew immediately to interview him, hoping that he would provide the answers they needed. They were disappointed. Taha told a story that seemed to confirm that Batouti had been confronted by the chief pilot, and he added some new details, but he turned out to be an informant of questionable utility—a radical Muslim who, along with others in the ranks of EgyptAir pilots, had forced the airline to ban the serving of alcohol, and who now went on at length about corruption at EgyptAir, and also what he claimed was rampant alcoholism and drug use among his secular peers. The request for asylum was itself a little flaky. The American investigators flew home without solid information. Most of this came out in the press when the story of Batouti's sexual improprieties was leaked, further angering the Egyptians. They countered, eventually producing a Boeing 777 captain

named Mohamed Badrawi, who had been with the other pilots in New York on the fateful night, and who testified at length that they were like a band of brothers—that Batouti and the chief pilot got along well and had had no direct confrontations. Rather, Badrawi said, he had acted at times as a "mediator" between the two men, cautioning Batouti on behalf of the chief pilot to "grow up" in order to avoid legal problems in the United States.

With that on the record, assigning a motive to Batouti became all the more difficult. For a variety of reasons, Bernard Loeb thought the FBI was wasting everyone's time. He did not really oppose the search for a motive, but he was against entering such speculative and easily countered discussions into the NTSB's public record. Privately he believed in the story of the fight. But as he later emphasized to me, "We just didn't need to go there."

Loeb thought the same about much of the investigation. Month after month, as the NTSB chased down the theories that EgyptAir kept proposing, Loeb worried about all the other projects that were being put aside. He tried to keep a sense of distance from the work, driving from suburban Maryland to his office dressed in a sports jacket and tie, just like any other Washingtonian with a quiet job. But it was a hopeless ambition. Most mornings the Egyptian delegation was there too. Later Loeb said to me, his voice strangled with frustration, "You had to be there! You had to live through this! Day in and day out! It was as if these people would go back to their rooms at night and then identify some kind of reason ... And then it would start all over again. It was insane! It was just insane!"

To bolster their arguments the Egyptians had hired some former accident investigators and also the retired NTSB chairman Carl Vogt, whose willingness to legitimize the Egyptian campaign was seen by many within the NTSB as a betrayal. The Egyptians also turned to the American pilots' union—in principle to improve their communication with the NTSB, but in practice probably just to add weight to their side. In the spring of 2000 the union sent to Washington a man named Jim Walters, a U.S. airline pilot with long experience in accident investigations. Walters thought he could patch things up. Later he said to me, "The Egyptians appeared to be listening to

me. But as it turned out, they weren't." Then he said, "I thought I was there to give them advice..." It was a disappointment. He liked the Egyptians personally, and remained sympathetic to their side even after he left.

I asked him to describe the scene in Washington. He said, "The NTSB isn't terribly tolerant of people who don't follow good investigative procedure. And they weren't used to dealing with a group like this, right in their back yard, with offices in the same building, there *every* day. I thought, The first thing we have to do is calm everybody down. I thought I could explain to the Egyptians, 'This is how the NTSB operates,' and explain to Jim Hall, 'Hey, these guys are Egyptians. You've got to understand who these guys are, and why they're doing things the way they are, and maybe we can all just kiss and make up and get along from here.'"

But it didn't work out that way. Walters was naive. Kiss and make up? The Egyptians no more needed his advice about investigative procedure than they had needed the NTSB's opinions about the nature of a free press.

A small war had broken out between Egypt and the United States on a battlefield called Loews L'Enfant Plaza Hotel. On one side stood Shaker Kelada and his men, fighting for the honor of their nation against the mysterious forces of American hegemony, and specifically against an agency whose famed independence they believed had been compromised. On the other side stood Bernard Loeb and his people, fighting just as hard—but to set a schedule, write the report, and disengage. Jim Hall was scurrying in between. And Boeing was off in Seattle, not quite out of range, trying unsuccessfully to look small.

The irony is that Loeb, too, thought the agency's independence had been compromised, though for the opposite reason: there were meetings at the White House, and phone calls to Jim Hall, in which concern was expressed about accommodating the Egyptian view, and in which it was implied that there should be no rush to finish a report that inevitably would offend Mubarak. Loeb was disgusted and typically vocal about his opinion. When I asked him if the influence was necessarily so wrong, he said, "Next they ask you to *change* the report—to say Batouti

didn't do it." He added, however, that no one had ever suggested such a change—and it was a good thing, too.

Egypt Versus the West

By late last May the fight had slowed, and Shaker Kelada was able to spend most of his time back home in Cairo. The NTSB had just issued a draft report, and Egypt was preparing an opposing response. I found Kelada in his expansive new office at the Cairo airport, where we talked several times over the course of a week. These were not good conversations. Kelada insisted on repeating the official Egyptian positions, and would go no further. At one point he began to attack the New York air-traffic controllers, and specifically Ann Brennan, for having walked away from her display. He implied that her absence had a bearing on the accident, or perhaps sparked a subsequent cover-up by the American government. He said, "It was very sloppy air-traffic control, and not what the U.S. wants to show. They're number one at everything, and they don't want *anyone* to know that they have a sloppy operation in New York."

I tried to reach him as one pilot to another. I said, "Come on, I think of that as being a normal operation, don't you?"

He said, "Well, if it is, I don't want to fly in the New York area!"

It was nonsense. And in aviation terms, a lot of what he said to me was equally unconvincing. Eventually I stopped taking notes.

Even when he was being reasonable, the party line kept showing through. He said, "I cannot say it's a mechanical failure. I don't have enough evidence, but I cannot dismiss the possibility of a mechanical failure ... If I want to be careful."

I said, "On the other hand, you *do* have enough evidence to dismiss the human factor?"

And he said, "Yes."

"To dismiss the intentional act?"

"Yes." He paused. He said, "We search for the truth."

It was late in the day. Kelada sat behind his desk—a man in a big office with jets outside, a smart man, a careful man. I thought of the question that had plagued me all along: not whether the Egyptians were right or wrong but whether they really believed their own words. Loeb had said to me, "Do they *believe* it? I believe they believe in fear."

I went downtown, to an old coffeehouse near the Nile, and spent a few hours with Hani Shukrallah, a columnist and one of the more thoughtful observers of the Egyptian scene. Shukrallah is a small, nervous man, and a heavy smoker. He said, "I know that as far as the Egyptian government was concerned, the point that this was *not* pilot error, and that the Egyptian pilot did *not* bring it down—this was decided before the investigation began. It had to do with Egypt's image in the outside world ... The government would have viewed this exactly as it would, for example, an Islamic terrorist act in Luxor—something that we should cover up. So it got politicized *immediately*. And this became an official line: You are out there to prove that EgyptAir is not responsible. It became a national duty. It was us versus the West. And all the history played into it, from Bonaparte's campaign until now." In the minds even of people on the street, Shukrallah said, it became "an all-out war."

Following Flight 990's Path

If so, the United States was in such a strong position that it could lose the struggle only by defeating itself. This is why from the very start of the difficult process it was all the more important for the NTSB to consider the evidence

fairly and keep an open mind. The problem was that so many of the scenarios the Egyptians posited were patently absurd—stray missiles, ghost airplanes, strange weather, and the like. Yet that didn't mean that everything they said was wrong. As long as Batouti's motive could not be conclusively shown, the possibility remained that the dive of Flight 990 was unintentional, just as Kelada maintained. And in the background the Egyptians had some very smart engineers looking into the various theories.

The 767's elevator movements are powered by three redundant hydraulic circuits, driving a total of six control mechanisms called "actuators," which normally operate in unison. Given the various linkages and cross-connections, the system is complex. The Egyptians thought it through and realized that if two of the six actuators were to fail on the same side of the airplane, they would drive both elevators down, forcing the 767 to pitch into a dive that might match the profile that had emerged from EgyptAir 990's flight-data recorder. Furthermore, if such a failure happened and either pilot tried to right it, that could conceivably explain the "splitting" of the elevators that occurred during 990's attempted recovery.

As might be expected, the discussion about dual actuator failures grew complicated. It also grew political. The NTSB had salvaged most of the actuators from the ocean floor and had found no clear evidence of failure, but with perceptions of public safety at stake, the agency asked Boeing for further information. Boeing engineers calculated that a dual actuator failure would not have deflected the elevators far enough down to equal the known elevator deflections of Flight 990, and that such a failure therefore would not have caused as steep a dive. To explore the question they performed a series of ground tests of a 767 elevator, inducing dual actuator failures and "splits" on a parked airplane in Seattle. After adjusting the measured effects for the theoretical aerodynamic pressures of flight, they found—as they had expected—poor correlation with the known record of Flight 990 elevator positions. They believed in any case that either pilot could quickly have recovered from a dual actuator failure by doing what comes naturally at such moments—pulling back hard on the controls.

The NTSB was satisfied; the Egyptians were not. They poked holes in the conclusions and requested basic and costly aerodynamic research, at speeds well beyond the 767's limits, toward Mach One. The question was, of course, To what end? But for Boeing this was a delicate thing, because Egypt kept buying expensive airplanes and was influential in the Arab world. A bit of additional research would perhaps be in order.

Meanwhile, the company's engineers had moved on to flight simulations of the accident, a series of dives set up to be flown in Boeing's highly programmable 767 engineering simulator—a "fixed cab" without motion, capable of handling extremes. These were the profiles that I flew when I went to Seattle last summer. On that same trip I went to Everett, Washington, where the airplanes are made, and in a cockpit with a company test pilot split the elevators in a powered-up 767, as the Egyptian crew presumably had. In order to do this we needed to break the connection between the left and right control yokes, which are mechanically joined under the floorboards, and usually move together. He pushed on his, I pulled on mine, and at fifty pounds of pressure between us the controls were suddenly no longer working in tandem. Far behind us, at the tail, the elevators separated smoothly. On a cockpit display we watched each elevator go its own way. The airplane shuddered from the movement of the heavy control surfaces. We played with variations. Toward the end the pilot laughed and said I was compressing his bones.

But when I got to the simulations, they felt too real to be a game. The simulator was a surrogate cockpit already in flight—humming and warm, with all the controls and familiar displays, and a view outside of an indistinct twilight. It was headed east at 33,000 feet and .79 Mach—just as Flight 990 had been. The first set of profiles were "back-driven" duplications of the fatal dive, generated directly from Flight 990's flight-data recorder. Another Boeing test pilot sat in Batouti's seat, and the engineers clustered around behind. I let the simulation run on automatic the first few times, resting one hand on the controls to feel the beast die—the sudden pitch and shockingly fast dive, the clicking of a wildly unwinding altimeter, the warbling alarm, the loss of most displays at the bottom after the engines were gone, and the dark, steep, soaring climb up to 24,000 feet, the control yoke rattling its warning of an aerodynamic stall, the airplane rolling southeast to its end. I watched this several times

and then flew the same thing by hand, matching the pressure I put on the control yoke to a specially rigged indicator, which, after the elevators' split had occurred, allowed me to match the force required to achieve Habashi's "pull" and Batouti's "push" as captured by the flight-data recorder. First I stood and flew Habashi's "Pull with me!" from behind the seat—up to ninety pounds of force, which under those conditions seemed like not very much. It was the other intention, the pushing, that was dramatic. What was required was not only pushing but then pushing harder. The idea that someone would do that in an airplane full of passengers shocked me as a pilot. If that's what Batouti did, I will never understand what was going on in his mind.

The second set of simulations were easier to fly. These were the dual actuator failures, which EgyptAir proposed might have overcome Batouti when he was alone in the cockpit. The purpose was to test the difficulty or ease of recovery from such an upset. Again the simulations began at 33,000 feet and .79 Mach. I flew by hand from the start. The airplane pitched down strongly and without warning. I hauled back on the controls and lost 800 feet. It was an easy recovery, but not fair—I had been ready. The engineers then made me wait before reacting, as they had made other pilots—requiring delays of five, ten, and finally fifteen seconds before I began the recovery. Fifteen seconds seems like an eternity in a 767 going out of control. Even so, by hauling hard on the yoke and throttling back, I managed to pull out after losing only 12,000 feet; and though I went to the maximum allowable dive speed, the airplane survived. This was not unusual. Airplanes are meant to be flown. During the original simulation sessions done for the NTSB every pilot with a dual actuator failure was able to recover, and probably better than I. So what was wrong with Batouti? The simplest explanation is that he was trying to crash the airplane. But if he wasn't, if the Egyptians were right that he couldn't recover from a dual actuator failure, what was wrong with him as an aviator?

I posed the question to Jim Walters, the airline pilot who despite his disappointment remained sympathetic to the Egyptians' position. He had a ready answer. He called Batouti "the world's worst airline pilot."

But how good do you have to be?

Bernard Loeb would have none of it. He said, "Sure. In the end they were willing to sell him down the river. They said, 'He panicked!' Bottom line is, if the actuator drops the nose, you can pull it up. They know that. They *admit* it. Pulling the nose up is the most intuitive, reflexive thing you can do in an airplane. So when you start hearing arguments like that, you *know* people are blowing smoke."

"Look, first we sit through this cockpit voice recording in which ... " He shook his head. "How many cockpit voice recordings have I heard? Hundreds? Thousands? When someone has a problem with an airplane, you know it. One of our investigators used to say to me, 'These damned pilots, they don't tell us what's happening. Why don't they say, "It's the rudder!"' They don't do that. But I'll tell you what they do say. They make clear as hell that there's something really wrong. 'What the *hell's* going on? What is *that*?' Every single one of them. When there's a control problem of some sort, it is so crystal clear that they are trying desperately to diagnose what is going on. Right to when the recorder quits. They are fighting for their lives.

"But this guy is sitting there saying the same thing in a slow, measured way, indicating no stress. The captain comes in and asks what's going on, and he doesn't answer! That's what you start with. Now you take the dual actuator failure that doesn't match the flight profile, and is also fully recoverable. Where do you want to go after that?"

The NTSB's final report on Flight 990 was expected for the fall of this year, and it was widely presumed in aviation circles that the report would find no mechanical failure or external cause for the crash. It also seemed likely that the report would at least implicitly blame Batouti for the disaster—a conclusion that would, of course, be unacceptable within Egypt. Nonetheless, by last May, when I met him in Cairo, Shaker Kelada was looking pleased, and I later found out why. His engineers had gotten busy again, and had come up with new concerns—certain combinations of tail-control failures that might require further testing. Now Boeing had come to town for a quiet talk with its

customers, and had agreed to do the tests. Boeing was going to inform the NTSB of the new work, and the end would again be delayed.

Sitting in his office, Kelada could not help gloating. He said, "Jim Hall told me, 'I've learned a very good lesson. When you deal with a foreign carrier in an investigation, before you go anywhere with it, you have to study the history and culture of the country.' These were his own words to me! He said, 'I knew nothing about Egypt or its culture before we got into EgyptAir 990.'"

I said, "What would he have learned?"

"Not to underestimate people. To think that he's way up there, and everybody's way down here."

Fair enough. But in the end there was the question of the objective truth—and there was the inclination not to seek real answers for even such a simple event as a single accident nearly two years before.

I knew that at the start of the investigation the Egyptian delegation had included a man named Mamdouh Heshmat, a high official in civil aviation. When the cockpit voice recording first arrived at L'Enfant Plaza, Heshmat was there, and he heard it through with a headset on. According to several investigators who listened alongside him, he came out of the room looking badly shaken, and made it clear he knew that Batouti had done something wrong. He may have called Cairo with that news. The next day he flew home, never to reappear in Washington. When NTSB investigators went to Cairo, they could not find him, though it was said that he was still working for the government. I knew I wouldn't find him either, but I wanted to see how Kelada would react to the mention of him. Kelada and I had come to the end. I said I had heard about a man who had been one of the first to listen to the tape—who could it have been? Kelada looked straight at me and said, "I don't recall his name." There was no reason to continue, from his perspective or mine.

ABOUT THE AUTHOR

William Langewiesche

William Langewiesche, a former national correspondent for *The Atlantic* and a professional pilot, has written about subjects including aviation, national security, and North Africa.

About



Contact



Podcasts



Subscription



Follow



[Privacy Policy](#) | [Your Privacy Choices](#)  | [Advertising Guidelines](#) | [Terms & Conditions](#) | [Terms of Sale](#)
| [Responsible Disclosure](#) | [Site Map](#)

TheAtlantic.com © 2026 The Atlantic Monthly Group. All Rights Reserved.

This site is protected by reCAPTCHA and the Google Privacy Policy and Terms of Service apply

The Atlantic